

Class nr. 9

Subject: 1. Consumer Theory

1.1. Axiom of Revealed Preferences (practice classes)

Exercise 1 from Varian's book (2010), p. 135

When prices are $(p_1, p_2) = (1, 2)$ a consumer demands $(x_1, x_2) = (1, 2)$, and when prices are $(q_1, q_2) = (2, 1)$, the consumer demands $(y_1, y_2) = (2, 1)$. Is this behavior consistent with the model of maximizing behavior?

Exercise 2 from Varian's book (2010), p. 135

When prices are $(p_1, p_2) = (2, 1)$ a consumer demands $(x_1, x_2) = (1, 2)$, and when prices are $(q_1, q_2) = (1, 2)$, the consumer demands $(y_1, y_2) = (2, 1)$. Is the consumer's behavior consistent with the model of maximizing behavior?

Exercise 3 from Varian's book (2010), p. 135

In the preceding exercise, which bundle is preferred by the consumer, the x-bundle or the y-bundle?

Exercise 7.1. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", pp. 80-1

When prices are $(4, 6)$, Goldie chooses the bundle $(6, 6)$ – observation A and when prices are $(6, 3)$, she chooses the bundle $(10, 0)$ – observation B.

- a) Sketch (in a graph) each of Goldie's budget constraints and choices.
- b) Is Goldie's behavior consistent with the Weak Axiom of Revealed Preference?

Exercise 7.4. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", p. 83

Here is a table of prices and the demands of a consumer named Ronald whose behavior was observed in 5 different price-income situations.

Situation	p_1	p_2	x_1	x_2
A	1	1	5	35
B	1	2	35	10
C	1	1	10	15
D	3	1	5	15
E	1	2	10	10

- Sketch (in a graph) each of Ronald's budget constraints and choices.
- Is Ronald's behavior consistent with the Weak Axiom of Revealed Preference?
- Shade lightly all of the points that you are certain **are worse** for Ronald than the bundle C.
- Suppose that you are told that Ronald has convex and monotonic preferences and that he obeys the Strong Axiom of Revealed Preference. Shade lightly all of the points that you are certain are **at least as good as** the bundle C.

Exercise 7.10. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", pp. 89-91

Consider the following tables with the chosen prices and quantities:

Prices	1830	1850	1890	1913
Grain Flour	0.14	0.14	0.16	0.19
Meat	0.28	0.34	0.66	0.85
Milk	0.07	0.08	0.10	0.13
Potatoes	0.032	0.044	0.051	0.064

Quantities	1850	1890
Grain Flour	165	220
Meat	22	42
Milk	120	180
Potatoes	200	200

- a) Complete the following table, which reports the annual cost of the 1850 and 1890 bundles of staple foods at various years' prices.

Cost	1850 bundle	1890 bundle
Cost at 1830 Prices	44.1	61.6
Cost at 1850 Prices		
Cost at 1890 Prices		
Cost at 1913 Prices	78.5	113.7

- b) Is the 1890 bundle revealed preferred to the 1850 bundle?
- c) Calculate the Laspeyres quantity index of staplefood consumption for 1890 with base year 1850.
- d) Calculate the Paasche quantity index for 1890 with base year 1850.
- e) Calculate the Laspeyres price index for 1890 with base year 1850.
- f) Calculate the ratio between the expenditure that would have to exist in 1850 to buy the 1890 bundle and the actual expenditure made in 1850.
- g) Calculate the ratio between the expenditure that would have to exist in 1890 to buy the 1850 bundle and the actual expenditure made in 1890.